

The Complete Linux Command Line Handbook

 [Linux](#)
 [Terminal](#)
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1. Core Navigation & File Management

Command	Description	Example Syntax
<code>pwd</code>	Print current working directory	<code>pwd</code>
<code>ls</code>	List directory contents	<code>ls -la</code> (<i>All files + detailed attributes</i>)
<code>cd</code>	Change current directory	<code>cd /var/www</code> or <code>cd ~</code> (<i>Home</i>)
<code>mkdir</code>	Create new directories	<code>mkdir -p project/src/components</code>
<code>rmdir</code>	Remove empty directory	<code>rmdir old_folder</code>
<code>touch</code>	Create an empty file or update timestamp	<code>touch server.js</code>
<code>cp</code>	Copy files or directories	<code>cp -r src/ dist/</code>
<code>mv</code>	Move or rename files/directories	<code>mv old_name.txt new_name.txt</code>
<code>rm</code>	Remove files or directories permanently	<code>rm -rf target_folder/</code>
<code>find</code>	Search files by name, type, size, date	<code>find . -name "*.log" -mtime -1</code>
<code>locate</code>	Fast file search using indexed	<code>locate nginx.conf</code>

Command	Description	Example Syntax
	database	
<code>ln</code>	Create hard or symbolic links	<code>ln -s /opt/app/current app</code>
<code>tree</code>	Display directory structure visually	<code>tree -L 2</code>
<code>clear</code>	Clear the terminal display	<code>clear</code> or press <code>Ctrl + L</code>



Pro-Tip: Always double-check path references when running `rm -rf` to prevent accidental deletion of critical system files!

2. File Inspection & Editing

```
# View file contents directly in terminal
cat application.log

# View large files page-by-page (Q to exit)
less /var/log/syslog

# Inspect the top 15 lines of a file
head -n 15 dataset.csv

# Monitor continuous logs in real-time
tail -f -n 50 /var/log/nginx/access.log

# Compare two files line by line
diff config.old.yml config.new.yml

# Count lines, words, and bytes in a file
wc -l access.log

# Check the type of a file
```

```
file archive.tar.gz

# Command-line text editors
nano config.env      # Beginner friendly
vim /etc/nginx.conf  # Advanced modal editor
```

3. Users, Groups & Permissions

Permission Matrix Cheat Sheet

- r (Read) = 4
- w (Write) = 2
- x (Execute) = 1

Commands

Command	Action	Example
<code>chmod</code>	Change file/folder permissions	<code>chmod 755 script.sh (rwxr-xr-x)</code>
<code>chown</code>	Change file owner and group	<code>sudo chown -R www-data:www-data /var/www</code>
<code>sudo</code>	Execute command with superuser privileges	<code>sudo systemctl restart nginx</code>
<code>useradd</code>	Add a new user account	<code>sudo useradd -m -s /bin/bash devuser</code>

Command	Action	Example
<code>passwd</code>	Set or update a user password	<code>sudo passwd devuser</code>
<code>usermod</code>	Modify user attributes / add to groups	<code>sudo usermod -aG docker devuser</code>
<code>groupadd</code>	Create a new group	<code>sudo groupadd deployers</code>
<code>whoami</code>	Print active logged-in user	<code>whoami</code>
<code>id</code>	Show UID, GID, and group memberships	<code>id devuser</code>
<code>su</code>	Switch to another user session	<code>su - devuser</code>

4. Process & System Monitoring

```
# Display active running processes filtered by name
ps aux | grep node

# Interactive real-time process monitoring
top

# Enhanced interactive process viewer (Requires
installation)
htop

# Terminate a process by Process ID (PID)
kill -9 10243
```

```
# Terminate all processes matching a string pattern
pkill -f "node server.js"

# Run a process in background detached from terminal
session
nohup python3 script.py > output.log 2>&1 &

# List background/foreground jobs in current shell
jobs

# Bring a background job to the foreground
fg %1

# Manage automated cron jobs
crontab -e

# Show system uptime and load average
uptime

# Display kernel, OS, and hardware info
uname -a
```

5. Disk Space & Storage

```
# Display filesystem disk space usage (Human readable
format)
df -h

# Summarize directory space consumption
du -sh /var/log/*

# Check RAM and Swap usage in gigabytes
free -h -g

# List block devices (disks, partitions)
lsblk

# Mount and unmount a filesystem
```

```
sudo mount /dev/sdb1 /mnt/data
sudo umount /mnt/data

# Check disk read/write activity in real time
iostat -x 2
```

6. Package Management

```
# Debian / Ubuntu (APT)
sudo apt update && sudo apt upgrade -y
sudo apt install nginx -y
sudo apt remove nginx --purge
apt list --installed

# RHEL / CentOS / Fedora (YUM / DNF)
sudo dnf install nginx -y
sudo dnf remove nginx
sudo dnf update -y

# Arch Linux (Pacman)
sudo pacman -Syu
sudo pacman -S docker

# Universal package formats
sudo snap install code --classic
flatpak install flathub org.videolan.VLC
```

7. Advanced Networking & Firewalls

```
# Show IP addresses and network interfaces
ip a

# Show routing table
ip route
```

```
# Test connectivity to a host
ping -c 4 google.com

# Trace the network path to a destination
traceroute github.com

# List open ports and listening services
sudo netstat -tulnp
sudo ss -tulnp

# Check DNS resolution
nslookup example.com
dig example.com +short

# Download a file from a URL
wget https://example.com/file.tar.gz
curl -O https://example.com/file.tar.gz

# UFW firewall (Ubuntu)
sudo ufw allow 22/tcp
sudo ufw allow 'Nginx Full'
sudo ufw enable
sudo ufw status verbose

# firewalld (RHEL/CentOS)
sudo firewall-cmd --add-port=8080/tcp --permanent
sudo firewall-cmd --reload

# iptables (low-level, legacy)
sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
```

8. Text Processing & Regex Mastery

```
# Search for a pattern inside files (recursive, line numbers)
grep -rn "ERROR" /var/log/app/
```



```
# Search with extended regex, case-insensitive
grep -Ei "fail(ed|ure)" server.log

# Stream editor: find and replace in-place
sed -i 's/localhost/127.0.0.1/g' config.yml

# Print specific columns from structured text
awk -F',' '{print $1, $3}' data.csv

# Sort and remove duplicate lines
sort names.txt | uniq -c | sort -nr

# Count occurrences of a word
grep -o "error" app.log | wc -l

# Extract text between patterns using sed
sed -n '/START/,/END/p' file.txt

# Cut specific fields by delimiter
cut -d':' -f1 /etc/passwd
```

9. Systemd Services & Logs

```
# Start, stop, restart a service
sudo systemctl start nginx
sudo systemctl stop nginx
sudo systemctl restart nginx

# Enable a service to launch on boot
sudo systemctl enable nginx

# Check current service status
systemctl status nginx

# View real-time logs for a specific service
journalctl -u nginx -f

# View logs since a specific time
```

```
journalctl --since "1 hour ago"

# List all active systemd units
systemctl list-units --type=service

# Reload systemd manager configuration
sudo systemctl daemon-reload
```

10. Compression, SSH & File Sync

```
# Create a compressed tar archive
tar -czvf backup.tar.gz /var/www/project

# Extract a tar.gz archive
tar -xzvf backup.tar.gz -C /destination/

# Zip and unzip files
zip -r project.zip project/
unzip project.zip

# Generate an SSH key pair
ssh-keygen -t ed25519 -C "you@example.com"

# Copy your public key to a remote server
ssh-copy-id user@192.168.1.10

# Connect to a remote server via SSH
ssh user@192.168.1.10 -p 22

# Securely copy files to/from a remote server
scp -r ./dist user@server:/var/www/app

# Sync directories efficiently (local or remote)
rsync -avz --progress ./project/
user@server:/var/www/project/
```

11. Shell Operators & Redirection

```
# Redirect stdout to a file (overwrite)
echo "Deploy complete" > status.txt

# Redirect stdout to a file (append)
echo "New log entry" >> status.txt

# Redirect stderr to a file
command 2> error.log

# Redirect both stdout and stderr to the same file
command > all_output.log 2>&1

# Pipe output of one command into another
ps aux | grep nginx | awk '{print $2}'

# Run commands sequentially regardless of success
command1 ; command2

# Run second command only if first succeeds
command1 && command2

# Run second command only if first fails
command1 || command2

# Run a command in the background
long_task.sh &

# Command substitution
current_date=$(date +%F)
```

12. Environment Variables & Shell Config

```
# Print all environment variables
printenv
```

```
# Print a specific variable
echo $PATH

# Set a temporary variable (current session only)
export API_KEY="your_key_here"

# Persist a variable across sessions
echo 'export API_KEY="your_key_here"' >> ~/.bashrc
source ~/.bashrc

# Create a command alias
alias ll='ls -la'
echo "alias ll='ls -la'" >> ~/.bashrc

# Check which shell you're using
echo $SHELL
```

13. DevOps Docker & Kubernetes Essentials

```
# --- Docker ---
docker build -t myapp:latest .
docker run -d -p 8080:80 --name myapp myapp:latest
docker ps -a
docker logs -f myapp
docker exec -it myapp /bin/bash
docker stop myapp && docker rm myapp
docker images
docker system prune -af

# Docker Compose
docker compose up -d
docker compose down
docker compose logs -f

# --- Kubernetes (kubectl) ---
kubectl get pods -A
kubectl describe pod mypod
kubectl logs -f mypod
```

```
kubectl apply -f deployment.yaml
kubectl delete -f deployment.yaml
kubectl exec -it mypod -- /bin/sh
kubectl get svc
kubectl scale deployment myapp --replicas=3
kubectl rollout restart deployment myapp
```

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